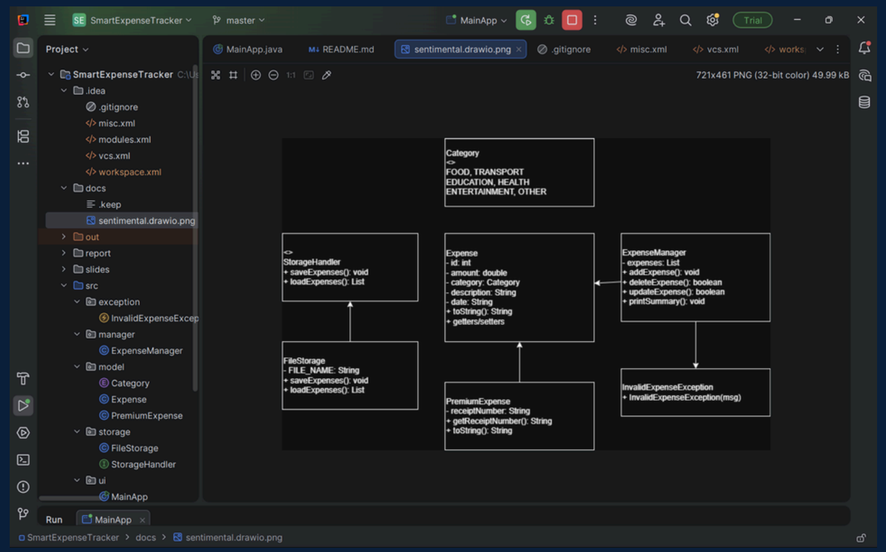
Tracker

Smart Expense

Presented by Deepsikha Halder

System Design and Package Structure



Package Model

Expense, Category and PremiumExpense classes with fields for amount, category, date and receipt number.

Package Manager

A single class that handles adding, viewing, updating and deleting expenses.

Package Storage

An interface and a file-based handler that saves and loads data from a simple CSV file.

OOP Concepts in Smart Expense Tracker

Encapsulation

Expense class manages data and behaviors securely.

Polymorphism

Same name, different results for toString() and Output shifts based on the object

Abstraction

StorageHandler interface simplifies storage operations effectively.

Inheritance

PremiumExpense inherits features from the Expense class.

Exception Handling

Stops negative amounts and Blocks empty descriptions

Technical Highlights of Smart Expense Tracker

File Storage

- Auto-saves all expenses
- Reloads everything when reopened

In-memory Management

- Stores expenses in a quick list
- Smooth for scanning and reports

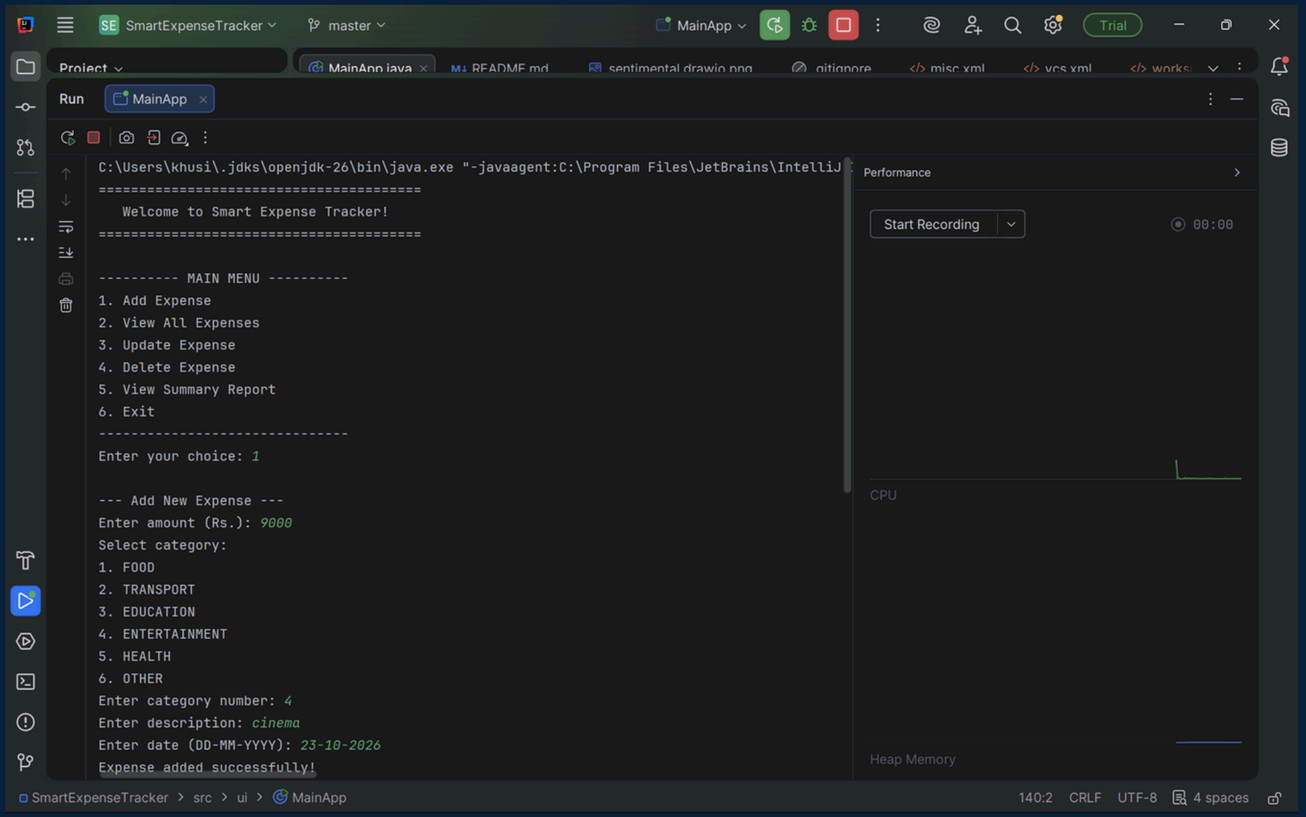
Input Validation

- Spots wrong amounts or missing text
- Shows helpful errors for easy fixes

Clean Architecture

- Each part has one clear job
- Ul stays separate from storage
- All actions run through ExpenseManager

Live Demo



Experience the **Smart Expense Tracker** through an interactive console menu. Users can seamlessly **add, view, delete**, and summarize their expenses, demonstrating the application's core functionality designed specifically for college students managing their finances effectively.